ClinicMate

A small language model for medical question answering in African healthcare settings

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Project links

GitHub repository: https://github.com/Tresorshingiro/ClinicMate

Demo video: **ClinicMate.mp4**

Live system: https://huggingface.co/spaces/Tresor26/ClinicMate

1. Abstract

Most medical question-answering benchmarks come from the United States, China and India, and say little about how language models handle the diseases, drugs and health systems of African countries. This project builds ClinicMate, an English-language medical question-answering system for African healthcare settings, using the AfriMed-QA dataset (Nimo et al., 2025). I fine-tuned Qwen2.5-1.5B-Instruct (Qwen Team, 2024) with QLoRA (Dettmers et al., 2023) on 3,663 multiple-choice (MCQ) and short-answer (SAQ) questions from my own 70/10/20 re-split of the dataset, and placed a DistilBERT classifier in front of it to refuse non-medical questions. On 739 held-out MCQs, fine-tuning raised accuracy from 49.7% to 54.1% (McNemar $p = 0.015$), against a majority-class baseline of 22.9%. On 243 SAQs, ROUGE-L rose from 0.156 to 0.292 and rescaled BERTScore from 0.132 to 0.302, although part of that gain comes from shorter answers that match the length of the references. A manual review of 50 answers found at least one hallucinated claim in half of them and potentially harmful advice in five. The domain classifier reached 99.0% accuracy on its own test split but refused only 17 of 30 hand-written non-medical questions. The system is deployed as a Gradio app on Hugging Face Spaces. Fine-tuned on a free GPU, the small model picks up the format and local framing of the task and gains a few points of accuracy, and its answers are still too unreliable to use as medical advice.

2. Introduction

Large language models can now pass medical licensing-style exams. Med-PaLM reached 67.6% on MedQA (Singhal et al., 2023) and Med-PaLM 2 reached 86.5% (Singhal et al., 2025). These results come from benchmarks built mostly from United States licensing exams (Jin et al., 2021), Indian entrance exams (Pal et al., 2022) and PubMed abstracts (Jin et al., 2019). They do not tell us how a model handles questions about malaria treatment in a district hospital, drug availability in a rural clinic, or the conditions that are common in a particular African country.

The World Health Organization (2006) estimated that Africa carries about 24% of the global burden of disease with about 3% of the world's health workers. Tools that answer health questions could help students and health workers there if they are accurate in this context, and AfriMed-QA was built to measure that. Its authors evaluated 30 language models and found that biomedical models did worse than general ones and that small models suitable for edge devices struggled to reach a passing score (Nimo et al., 2025). Small models are still attractive in low-resource settings because they are cheap to run and can be hosted without paid hardware, so it is worth asking how far one can be pushed with the dataset's own training questions.

The task is medical question answering in English. The choice of language follows from the data: AfriMed-QA is described by its authors as the first large-scale pan-African, multi-specialty medical question-answering dataset, and it is in English, written by students and clinicians from over 60 medical schools in 16 African countries (Nimo et al., 2025). The questions refer to endemic diseases, locally available drugs and the constraints of resource-limited health systems. A general English model can read them without knowing this local context, and fine-tuning on the dataset is meant to supply it.

The intended users are medical and nursing students and health workers in African countries who want to check their knowledge against questions written by their own peers and teachers. Most of AfriMed-QA consists of exam-style questions from medical schools, which matches this group. Members of the public asking about their own health are a secondary audience: the app accepts their

questions, but Section 7 shows that the model is weakest on lay-language questions, and every answer carries a notice that it is not medical advice.

The objective of the project is to build and evaluate a small, deployable medical question-answering system for African healthcare settings. I set out to answer four questions. Does QLoRA fine-tuning on AfriMed-QA improve a small open model over its zero-shot performance? How much do model size, LoRA rank, training data and retrieval change the result? Can a lightweight classifier stop the system from answering non-medical questions? And what kinds of errors remain once the system is running?

3. Related Work

3.1 Medical question-answering benchmarks

MedQA collects multiple-choice questions from medical licensing exams in the United States, mainland China and Taiwan (Jin et al., 2021). MedMCQA contains about 194,000 questions from Indian postgraduate entrance exams covering 21 subjects (Pal et al., 2022). PubMedQA asks yes/no/maybe questions about PubMed abstracts (Jin et al., 2019). Singhal et al. (2023) combined these and other datasets into MultiMedQA and added human evaluation of long-form answers, because accuracy on exam questions does not capture whether an answer is safe or complete. AfriMed-QA follows the same design but draws its 15,000 questions from African medical schools and crowd workers, and includes consumer-style queries alongside expert MCQs and SAQs (Nimo et al., 2025).

3.2 Adapting language models to medicine

One line of work adapts models to biomedical text through continued pre-training, as in BioBERT (Lee et al., 2020) and MEDITRON (Chen et al., 2023). Another relies on very large general models with careful prompting and instruction tuning, as in Med-PaLM (Singhal et al., 2023). Both are expensive. Parameter-efficient fine-tuning makes adaptation affordable: LoRA freezes the pre-trained weights and trains low-rank update matrices in each layer (Hu et al., 2022), and QLoRA stores the frozen base model in 4-bit NormalFloat precision so that models with billions of parameters can be fine-tuned on a single GPU (Dettmers et al., 2023). Retrieval-augmented generation (Lewis et al., 2020) offers a third option, giving the model relevant passages at inference time instead of changing its weights.

3.3 Out-of-domain questions

A model fine-tuned only on medical answers has never seen a refusal, so it will try to answer anything. Hendrycks and Gimpel (2017) showed that softmax confidence gives a simple baseline for detecting out-of-distribution inputs. A common practical alternative is a separate binary classifier that decides whether a request is in scope before the main model runs. I use this approach with DistilBERT (Sanh et al., 2019), a smaller distilled version of BERT (Devlin et al., 2019).

3.4 The gap this project addresses

The AfriMed-QA paper benchmarks existing models on the dataset. It leaves open whether a small open model, fine-tuned with the dataset's own training questions on free hardware, can close part of the gap it found for small models, and what such a model needs around it to be used safely by the public. This project tests that directly, measures the effect of size, rank, training split and retrieval, and deploys the result behind a domain classifier.

4. Dataset and Data Preparation

4.1 Source and characteristics

I used AfriMed-QA v2, distributed on Hugging Face as intronhealth/afrimedqa_v2 under a CC-BY-SA-4.0 licence (Nimo et al., 2025). The file contains 15,275 rows and 29 columns. Table 1 shows the three question types and how they fall across the official split.

Table 1. AfriMed-QA v2 by question type, tier and official split

Question type	Tier	Official train	Official test	Gold answer
Consumer query	Crowdsourced	8,366	1,634	No (1.7% have one)
MCQ	Crowdsourced	129	0	Correct option(s)
MCQ	Expert	0	3,910	Correct option(s)
SAQ	Crowdsourced	877	0	Free-text answer
SAQ	Expert	0	359	Free-text answer

MCQs have up to five options. Most (3,122) use all five, 898 have four, and a few have two or three. 311 MCQs have more than one correct answer, and only 7.6% come with a written explanation. SAQ reference answers have a median length of 29 words, with a long tail up to 900 words. The 10,000 consumer queries are lay-language questions written from a patient's point of view, but only 1.7% of them have an answer. Question writers are concentrated in a few countries: Nigeria contributes 7,577 rows, followed by Kenya (2,476), South Africa (2,444) and Ghana (1,572). For single-answer MCQs the correct letter is spread fairly evenly over A to D (20.8% to 23.7%), with E rarer (11.6%) because many questions have only four options.

4.2 Challenges

Three features of the data shaped the design. First, the official split follows the tier: the training split is entirely crowdsourced and the test split entirely expert-written, with only 129 training MCQs. That is too few to fine-tune on, and it confounds the split with question difficulty. Second, consumer queries have no gold answers, so they cannot be used to train or score the generator. Third, the raw text has formatting problems, including broken bullet characters, escaped newlines, options stored as JSON-like strings with empty slots, and 765 consumer-query rows that duplicate another row.

4.3 Preprocessing

I wrote the cleaning code once in src/data.py and used it in every notebook. Text is normalised with Unicode NFKC, broken bullet characters are replaced, escaped and Windows line breaks are converted, and repeated whitespace is collapsed. Answer options are parsed from their JSON-like strings, empty or "N/A" options are dropped, and the remaining options are mapped to letters A to E. The correct answer, stored as option keys such as option4, is mapped to the matching letters. Consumer queries are stored as a template ending in "Question:", which I stripped to keep only the scenario as context.

Filtering then removed 25 rows with questions shorter than 10 characters, MCQs with fewer than two options or no correct answer, and SAQs with answers shorter than three words. This left 5,235 MCQs and SAQs with gold answers, and 5,234 after removing one exact duplicate. Consumer queries were deduplicated separately, leaving 9,453.

4.4 Splits

Because the official split cannot support training, I re-split the 5,234 MCQs and SAQs 70/10/20, stratified by question type and tier, giving 3,663 training, 523 validation and 1,048 test examples. The evaluation set is the test SAQs plus the single-answer test MCQs: 739 MCQs and 243 SAQs. Multi-answer MCQs stay in training but are left out of evaluation so that accuracy has a single correct letter. I kept the official split for one experiment (E4), training on 903 crowdsourced examples (101 held out for validation) and testing on 3,930 expert questions (3,589 MCQs and 341 SAQs). Results on my re-split are not comparable with the AfriMed-QA leaderboard.

The dataset suits the task because it is written by people working in African health systems and covers 32 specialties (Nimo et al., 2025). Its MCQs allow exact accuracy measurement, its SAQs test free-text generation, and its consumer queries supply real lay-language questions for testing the domain classifier.

4.5 Data for the domain classifier

The domain classifier needs medical and non-medical questions. The medical side uses the MCQ and SAQ questions from each split plus the consumer queries, which I split 70/10/20 in the same way. For version 1 of the classifier, the non-medical side came from SQuAD (Rajpurkar et al., 2016) and WebQuestions (Berant et al., 2013), with any question containing a medical keyword removed by a regular expression, leaving 87,414 questions. Version 2 replaced half of these with question titles from Yahoo Answers (Zhang et al., 2015), excluding the Health topic and filtering out medical and everyday health words such as pill, sleep or pregnant. Both versions have 20,560 training, 2,936 validation and 5,878 test examples, balanced 50/50. I also wrote a challenge set of 60 questions by hand, 30 medical and 30 non-medical, designed to be hard: casual symptom questions on one side and non-medical questions that use medical words ("how do I remove a virus from my laptop") on the other. It was never used for training.

5. Methodology

5.1 Prompt and target format

Every example becomes a chat with a system prompt, a user message and an assistant answer, formatted with the model's own chat template. The system prompt is the same for training, evaluation and deployment:

> "You are ClinicMate, a medical question-answering assistant focused on healthcare in African settings. Answer accurately and concisely using established clinical knowledge and consider resource-limited settings and diseases common in Africa. You do not replace a health professional."

For an MCQ, the user message lists the question and the lettered options and asks for "Answer: <letter>" followed by a short explanation. The training target is "Answer: B", followed by the explanation when one exists (truncated to 1,200 characters at a sentence boundary). For an SAQ, the user message is the question, preceded by its context when there is one, and the target is the reference answer truncated to 1,500 characters. During training, the loss is computed only on the target tokens: prompt tokens get the label -100 so the model is not trained to reproduce the question. The longest training example has 486 tokens (median 155, 95th percentile 248), so a maximum length of 1,024 tokens never truncates anything.

5.2 Baselines

B0 gives the floor for MCQs: a random letter from the available options, and always choosing C, the most common letter in the training set. B1 is a retrieval baseline for SAQs: it finds the most similar training question by TF-IDF cosine similarity over word unigrams and bigrams and returns that question's reference answer. B2 and B3 are Qwen2.5-0.5B-Instruct and Qwen2.5-1.5B-Instruct used zero-shot with the same prompt as the fine-tuned models.

B0 shows what accuracy is possible with no knowledge at all, which matters because the answer letters are not uniformly distributed. B1 shows how far SAQ metrics can be pushed by copying a real answer to a similar question, without generating anything. B2 and B3 are the most important comparison: they use the same models and the same prompts as the fine-tuned systems, so any difference between B3 and E2 comes from fine-tuning alone.

5.3 Fine-tuned generator

The generator is Qwen2.5-1.5B-Instruct (Qwen Team, 2024), released under Apache 2.0. It is small enough to fine-tune on one free Kaggle T4 GPU and to serve without paid hardware, and it already follows instructions.

Qwen2.5-1.5B is a decoder-only Transformer (Vaswani et al., 2017; Qwen Team, 2024). Input text is split into tokens by a byte-level BPE tokenizer with a vocabulary of 151,936 entries, and each token is mapped to a learned embedding vector of size 1,536. The embeddings pass through 28 Transformer blocks. Each block has a self-attention layer with 12 query heads that share 2 key-value heads (grouped-query attention), rotary position embeddings to encode word order, and a feed-forward layer with SwiGLU activation and an inner size of 8,960, with RMSNorm before each sub-layer. Attention is causal, so each position can only attend to earlier tokens. A final linear layer turns the last hidden state into a probability for every token in the vocabulary, and the answer is produced one token at a time. The input is the chat-formatted prompt from Section 5.1 and the output is the generated answer.

I fine-tuned it with QLoRA (Dettmers et al., 2023): the base weights are loaded in 4-bit NF4 with double quantisation and fp16 computation, and LoRA adapters (Hu et al., 2022) are added to all seven linear projections in every transformer block (query, key, value, output, gate, up and down). Dettmers et al. (2023) found that adapting all linear layers is needed to match full fine-tuning. With rank r = 16 and scaling factor alpha = 32 this trains 18.5 million parameters, about 1.2% of the model.

Training used the Hugging Face Transformers Trainer (Wolf et al., 2020) and the PEFT library (Mangrulkar et al., 2022) with a learning rate of 2e-4, cosine decay, 3% warm-up, dropout 0.05, an effective batch size of 16 (4 examples with 4 gradient-accumulation steps), fp16 mixed precision and gradient checkpointing, for three epochs. Validation loss was computed every 50 steps and the checkpoint with the lowest validation loss was kept. Training the 1.5B model took about 50 minutes on a T4.

MCQ predictions come straight from the model's next-token scores. The prompt ends with "Answer:", and the logits for the letters A to E (restricted to the letters that have options) are compared; the highest one is the prediction. This removes parsing errors and gives the same treatment to zero-shot and fine-tuned models. SAQ answers are generated greedily with at most 256 new tokens and a repetition penalty of 1.1.

5.4 Experiments

Holding everything else fixed, I varied one factor at a time. E1 fine-tunes the 0.5B model with r = 16 to measure the effect of model size. E2 is the main model, 1.5B with r = 16. E3 halves the rank to r =

8 (9.2 million trainable parameters). E4 trains the 1.5B model on the official crowdsourced split and tests on the expert questions. E5 adds retrieval to E2 at inference time: the two training questions closest to the test question, found with all-MiniLM-L6-v2 sentence embeddings (Reimers & Gurevych, 2019; Wang et al., 2020), are placed in the prompt with their answers truncated to 500 characters.

5.5 Domain classifier

I compared two classifiers. Gate A is TF-IDF over word unigrams and bigrams with logistic regression (Pedregosa et al., 2011). Gate B is distilbert-base-uncased (Sanh et al., 2019) fine-tuned for two epochs with learning rate 3e-5 and batch size 32. DistilBERT is a Transformer encoder with 6 layers, hidden size 768 and about 66 million parameters. Unlike the generator, it reads the whole question in both directions at once, and a small classification head on the hidden state of the first token outputs two scores, one for medical and one for out of domain. Gate A sees only which words occur, while Gate B uses contextual embeddings, so comparing them shows whether context helps. Each was trained twice, once on the version 1 data and once on version 2. In the deployed system, a question is refused when the classifier's probability that it is out of domain is above 0.5, and the generator is never called for a refused question.

5.6 Evaluation

Accuracy suits MCQs because each evaluated question has exactly one correct letter. ROUGE-L measures the longest common word sequence between an answer and the reference, so it rewards using the same facts in the same words. BERTScore compares contextual embeddings token by token, so it gives credit for paraphrases that ROUGE-L misses. Both depend on a single reference answer, and neither can tell whether a medical claim is true, which is why hand ratings were added. For the domain classifier the two kinds of error have different costs, so alongside accuracy I report the share of non-medical questions refused and the share of medical questions wrongly refused.

MCQ accuracy is reported with a 95% confidence interval from 2,000 bootstrap resamples (Efron & Tibshirani, 1993). Paired comparisons between two systems on the same MCQs use McNemar's test with continuity correction (McNemar, 1947), which looks only at questions where the systems disagree. SAQ answers are scored with ROUGE-L (Lin, 2004) and BERTScore F1 using roberta-large (Zhang et al., 2020; Liu et al., 2019). BERTScore is rescaled against its baseline, so unrelated text scores near zero and can go negative. Because neither metric checks clinical correctness, I also rated 50 answers from the main model by hand (30 test SAQs and 20 consumer queries) on the axes AfriMed-QA uses for its own human evaluation: correctness from 1 to 5, omission, hallucination, potential harm, and relevance to African settings from 1 to 5 (Nimo et al., 2025). I did the ratings myself and I have no medical training, which I return to in Section 9.

6. Experiments and Results

6.1 Main results

Table 2. Results on the re-split evaluation set (739 MCQs, 243 SAQs)

System	Trainable params	MCQ accuracy [95% CI]	Expert / crowd MCQ	SAQ ROUGE-L	SAQ BERTScore	SAQ words
B0 random	0	0.212 [0.183, 0.242]	–	–	–	–
B0 majority (C)	0	0.229 [0.199, 0.260]	–	–	–	–

System	Trainable params	MCQ accuracy [95% CI]	Expert / crowd MCQ	SAQ ROUGE-L	SAQ BERTScore	SAQ words
B1 TF-IDF retrieval	0	–	–	0.258	0.209	–
B2 0.5B zero-shot	0	0.316 [0.283, 0.350]	0.315 / 0.360	0.173	0.142	104
B3 1.5B zero-shot	0	0.497 [0.460, 0.533]	0.488 / 0.760	0.156	0.132	137
E1 0.5B QLoRA r=16	8.8M	0.400 [0.365, 0.437]	0.393 / 0.600	0.297	0.305	38
E2 1.5B QLoRA r=16	18.5M	0.541 [0.506, 0.578]	0.536 / 0.680	0.292	0.302	42
E3 1.5B QLoRA r=8	9.2M	0.548 [0.513, 0.586]	0.543 / 0.680	0.302	0.316	37
E5 E2 + retrieval	18.5M	0.525 [0.488, 0.562]	0.519 / 0.680	0.323	0.339	42
E2 merged, fp16	–	0.549 [0.516, 0.587]	0.543 / 0.720	0.312	0.334	38

Note. The crowdsourced group has only 25 MCQs, so its accuracy is very noisy. "E2 merged" is the deployed model: the E2 adapter merged into fp16 weights and evaluated without 4-bit quantisation.

Fine-tuning helped the 1.5B model on MCQs. Accuracy rose from 49.7% to 54.1%, and McNemar's test on the 739 paired answers gives $p = 0.015$ (Table 3). Fine-tuning fixed 98 questions that the zero-shot model got wrong and broke 66 that it had got right, a net gain of 32. The improvement is modest compared with the gap to the majority baseline (22.9%), which shows that most of the model's MCQ ability comes from pre-training.

Table 3. McNemar tests on paired MCQ predictions

Comparison	Only A correct	Only B correct	Both correct	p-value
A = B3 zero-shot, B = E2	66	98	302	0.015
A = E1 (0.5B), B = E2 (1.5B)	74	178	222	< 0.001
A = E2 (r=16), B = E3 (r=8)	25	30	375	0.59

Model size mattered far more than anything else I varied. The fine-tuned 0.5B model reached only 40.0%, and E2 beat it on 178 questions while losing on 74 ($p < 0.001$). Fine-tuning gave the 0.5B model a larger jump (8.4 points over its zero-shot score) but it still finished well below the zero-shot 1.5B model. This fits the AfriMed-QA finding that small models struggle on the benchmark (Nimo et al., 2025).

LoRA rank made no measurable difference. E3, with half the trainable parameters, scored 54.8% against 54.1% for E2, and the two disagreed on only 55 questions ($p = 0.59$). For a training set of this size, rank 8 is enough. I deployed E2 because it was the planned main model and was finished first; E3 would have served equally well.

6.2 Short-answer questions

On SAQs, every fine-tuned model roughly doubled ROUGE-L over its zero-shot version, and rescaled BERTScore rose from about 0.13 to about 0.30. Much of this comes from length. The zero-shot 1.5B model wrote 137 words on average and about 39% of its answers looked cut off at the 256-token limit, while E2 wrote 42 words, close to the references' median of 29. Across E2's answers, the absolute difference between predicted and reference length correlates with BERTScore at Spearman rho = -0.42. The TF-IDF retrieval baseline, which returns a real answer to a different question, scores higher than both zero-shot models (ROUGE-L 0.258). The metrics reward answers written in the style and at the length of the references, so most of the SAQ gain shows that the model learned how AfriMed-QA answers are written. Whether those answers are medically correct is a separate question, which the human evaluation in Section 7 takes up.

6.3 Retrieval, the official split and the deployed model

Retrieval (E5) gave the best SAQ scores (ROUGE-L 0.323, BERTScore 0.339) but lowered MCQ accuracy to 52.5%. The model was never trained with retrieved passages in its prompt, so at test time it sees an input format it has not learned. I did not measure how often the retrieved questions were relevant to the test question. Training with retrieved context would be needed for a fair test of retrieval.

The official split (E4) shows how hard the move from crowdsourced to expert questions is. Trained on 903 crowdsourced examples, the model scored 48.4% on 3,589 expert MCQs against 47.4% zero-shot, with overlapping confidence intervals ([0.469, 0.501] and [0.458, 0.492]). SAQ BERTScore stayed below zero for both. The tier gap is also visible in Table 2: the zero-shot 1.5B model answers 76% of crowdsourced MCQs correctly but only 48.8% of expert ones.

Merging the adapter into fp16 weights for deployment did not hurt. The merged model scored 54.9% on MCQs and slightly higher SAQ scores than the 4-bit evaluation of E2, probably because the merged model runs without the quantisation error of the 4-bit base.

6.4 Training behaviour

Figure 1 shows the training and validation loss for E2. Validation loss was lowest at step 250 (0.918), a little after the end of the first epoch (about 230 steps). It then flattened and jumped at the start of the third epoch while training loss kept falling to 0.24. This is overfitting: the model starts memorising the training answers. E3 behaved the same way (best at step 250, 0.913), E1 reached its best later (step 450), and E4 at step 120 of 171. Keeping the best checkpoint protected the results, but one or two epochs would have given the same models in less time.

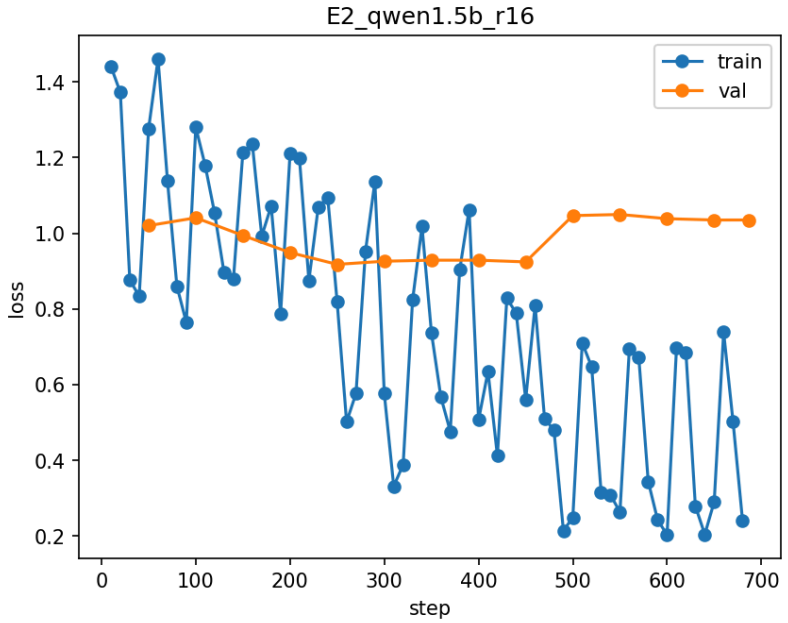


Figure 1. Training and validation loss for E2 (Qwen2.5-1.5B, QLoRA r = 16). The checkpoint at step 250 was kept.

6.5 Decoding settings

Before deployment I compared three decoding settings on the 243 test SAQs with the merged model (Table 4). I counted an answer as looping when fewer than 70% of its word 4-grams were unique. Blocking repeated 4-grams removed the two looping answers but lowered ROUGE-L, and a stronger

repetition penalty doubled the answer length and lowered both scores. I kept a repetition penalty of 1.1 for the app.

Table 4. Decoding settings for the merged model on 243 SAQs

Setting	ROUGE-L	BERTScore	Looping answers	Average words
Repetition penalty 1.1 (used)	0.312	0.334	0.8%	37.9
Penalty 1.1 + no repeated 4-grams	0.284	0.308	0%	37.8
Repetition penalty 1.2	0.226	0.232	0%	68.7

6.6 Domain classifier

Table 5. Domain classifier results

Classifier	Test v1 accuracy	Test v2 accuracy	Challenge accuracy	Challenge OOD refused	Challenge medical refused
A: TF-IDF + LR, v1 data	0.984	0.869	0.700	53.3%	13.3%
B: DistilBERT, v1 data	0.995	0.919	0.650	30.0%	0.0%
A: TF-IDF + LR, v2 data	0.976	0.972	0.833	83.3%	16.7%
B: DistilBERT, v2 data (used)	0.992	0.990	0.783	56.7%	0.0%

Note. The challenge set has 30 medical and 30 non-medical questions written by hand.

Both version 1 classifiers scored almost perfectly on their own test split and poorly on the challenge set. The version 1 DistilBERT model refused only 9 of the 30 non-medical challenge questions. The cause was the training data: the medical side was conversational and the non-medical side was formal trivia from SQuAD and WebQuestions, so the classifier learned to separate writing styles. Adding casual Yahoo Answers questions in version 2 made the non-medical side conversational too, and the DistilBERT model's refusal rate on the challenge set rose to 56.7% while it still accepted every medical challenge question. The TF-IDF model refused more non-medical questions (83.3%) but also refused 5 of the 30 medical ones, including "why do I feel tired all the time". I deployed the DistilBERT version 2 model because refusing a real health question is a worse failure for this application than answering an off-topic one. The challenge set guided the decision to build version 2, so its numbers for version 2 are probably optimistic.

7. Error Analysis and Discussion

7.1 Multiple-choice errors

Figure 2 breaks MCQ accuracy down by specialty for the zero-shot and fine-tuned 1.5B models, for specialties with at least 10 test questions. Rheumatology was the strongest (84%, n = 19) and haematology the weakest (33%, n = 15), where fine-tuning lowered accuracy from 47%. Fine-tuning helped most in urology, rheumatology and pulmonary medicine and slightly hurt neurology, nephrology and pathology. With 11 to 116 questions per specialty, most of these differences are within noise.

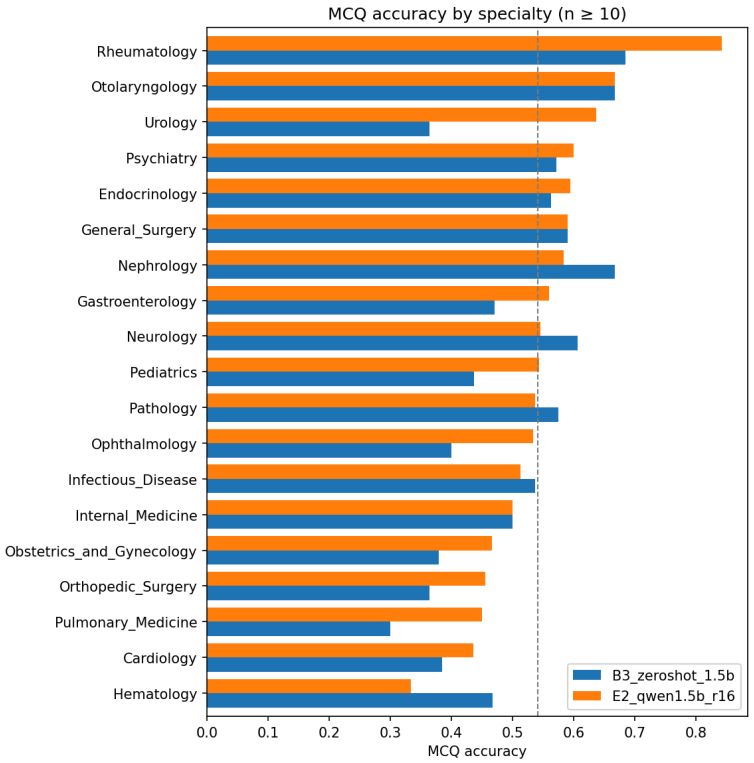


Figure 2. MCQ accuracy by specialty for the zero-shot (B3) and fine-tuned (E2) 1.5B models, specialties with at least 10 test questions. The dashed line is E2's overall accuracy.

By the question writer's country, the fine-tuned model scored 50.2% on Ghanaian questions (n = 281), 54.7% on Nigerian (n = 274), 57.5% on Kenyan (n = 106) and 61.2% on Malawian (n = 67). Ghanaian questions gained almost nothing from fine-tuning (49.8% to 50.2%). Longer questions, which are usually clinical vignettes, were not harder: the longest quarter (over 24 words) scored 57.8% and the shortest quarter 51.7%. Short recall questions such as "Drug lupus-like syndrome can be due to one of the following" depend on a single remembered fact, which a small model often lacks.

Language models are known to prefer some answer positions regardless of content (Zheng et al., 2024). Figure 3 compares the predicted letters with the gold letters. The zero-shot model over-predicted A (27.5% of predictions against 20.8% of answers), and fine-tuning removed most of that bias, leaving a smaller preference for D (26.1% against 24.6%).

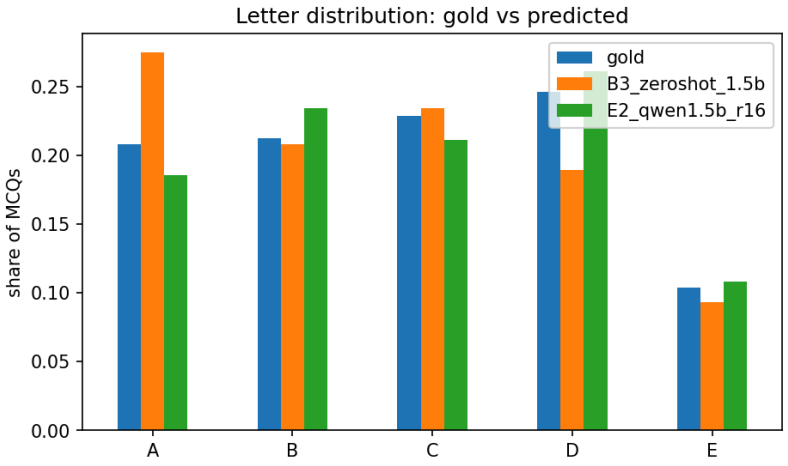


Figure 3. Share of MCQs by gold letter and by letter predicted by the zero-shot (B3) and fine-tuned (E2) 1.5B models.

One question that fine-tuning broke was "Hepatic tuberculosis may manifest as all of the following except": the zero-shot model correctly chose E (Rich focus) and the fine-tuned model chose A (miliary tuberculosis). Two of the five broken examples I inspected use "except" or "all of the above", where the answer depends on checking every option. In another, a man with chronic alcohol use and a hard, scalloped liver, the zero-shot model correctly chose alpha-fetoprotein while the fine-tuned model chose a garbled option, "C.19-19", probably because it resembles the familiar marker CA 19-9.

7.2 Short-answer errors

I read the 25 SAQ answers with the lowest BERTScore and labelled each with one error category (Figure 4). Nine were verbosity or truncation problems. These are mostly repetition loops: one answer about investigations after surgery starts with sensible tests and then repeats "LFTs" about 60 times, and an answer about hyperkalaemia is a list from a) to v) where every item is "N/a". Another nine were factual errors, four were correct but incomplete, two had a poor reference answer, and one was correct but penalised by the metric: for tumour markers of pancreatic cancer, the model listed CA 19-9 first, which matches the reference, and then padded the list with six non-specific markers.

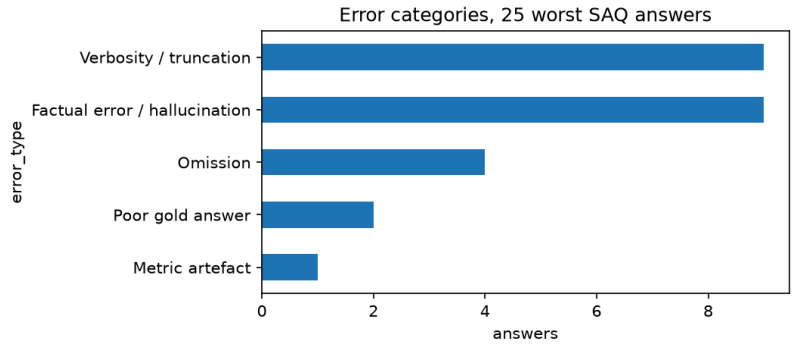


Figure 4. Error categories among the 25 lowest-scoring SAQ answers from E2, labelled by hand.

7.3 Human evaluation

Table 6. Human ratings of 50 answers from the fine-tuned model (E2)

Group	Correctness (1 to 5)	African context (1 to 5)	Omission	Hallucination	Potentially harmful	Rated 4 or 5
SAQ (n = 30)	3.30	2.87	60%	43%	7% (2)	50%
Consumer query (n = 20)	2.35	1.20	85%	60%	15% (3)	20%
All (n = 50)	2.92	2.20	70%	50%	10% (5)	38%

Note. Rated by the author, who has no medical training.

The best answers were short and specific to the setting. Asked about barriers to prenatal care in rural African communities, the model named transport, cost, cultural beliefs about childbirth and the shortage of trained midwives and doctors. Asked about mental health management in African communities, it described stigma, reluctance to seek treatment and reliance on traditional and spiritual healers, which matches the reference.

The worst answers stated wrong facts in the same confident tone. Asked for the first-line treatment of uncomplicated syphilis, the model said tetracycline or doxycycline; the first-line drug is benzathine penicillin G, and doxycycline is an alternative that should not be used in pregnancy. For acute bacterial prostatitis it suggested metronidazole or nitrofurantoin, neither of which is appropriate. It said haemoglobin electrophoresis diagnoses parvovirus B19 infection, when the test detects

haemoglobin variants. These are typical hallucinations, fluent statements that no source supports (Ji et al., 2023). Because they are written in the same form as correct answers, a reader without medical training has no way to spot them.

Consumer queries were the weakest group, with an average correctness of 2.35 and almost no African context (1.20). The model was never trained on lay-language questions with answers. Asked by a patient whether their treatment would affect urination, it gave false reassurance and invented a bladder-cancer drug called "Bleocin". Asked about the complications of leishmaniasis, it described leprosy. For one input that was a statement rather than a question, it answered "Answer: N/a", another trace of the MCQ format.

AfriMed-QA reported that small models struggle on the benchmark (Nimo et al., 2025), and the 1.5B model here, even after fine-tuning, answers about half of the expert MCQs wrongly and hallucinates in half of the reviewed free-text answers. Singhal et al. (2023) argued that automatic metrics and exam accuracy are not enough to judge medical answers, and here the SAQ metrics improved sharply while hand review showed many answers that were well formed but clinically wrong.

7.4 Out-of-domain behaviour

To show why the domain classifier is needed, I sent 15 of the non-medical challenge questions straight to the fine-tuned model. It answered every one. Some answers were correct ("Germany won the 2014 World Cup", "Chinua Achebe wrote Things Fall Apart"), some were plausible instructions (a recipe for jollof rice) and some were invented ("1 Naira = 350 Kobo", when 1 naira is 100 kobo, and a start time for "the football match tonight"). Fine-tuning on medical answers did not teach the model to stay in its domain.

The classifier's remaining mistakes are non-medical questions that use medical vocabulary: "how do I remove a virus from my laptop", "how much does it cost to build a hospital in Ghana", "how do I treat a sick tomato plant", "my cow has a cough what should I do" and "which university has the best medical school in Nigeria". The classifier still relies on words such as virus, hospital, cough and medical more than on what the question is about. Harder negative examples of this kind in the training data would be the most direct fix.

8. Deployment

For deployment, the E2 adapter was merged into the base model's weights in fp16 and pushed to the Hugging Face Hub with a model card that states the training data, the evaluation results, the known limitations and that the model is not for clinical use. The version 2 DistilBERT classifier was pushed alongside it.

The interface is a Gradio app (Abid et al., 2019) hosted on Hugging Face Spaces. A user types a health question and presses Ask or Enter. The app first runs the domain classifier on the CPU. If the probability that the question is out of domain is above 0.5, it returns a polite refusal together with that probability, and the generator is not called. Otherwise it builds the same system prompt and "Question: ..." format used for SAQ training and generates an answer with the settings from Section 6.5 (greedy decoding, repetition penalty 1.1, at most 256 new tokens). The answer is streamed to the page as it is generated, and every answer ends with a notice that ClinicMate is a student project, not medical advice, and that emergencies should go to the nearest health facility. Four example questions are provided, including one about the World Cup to demonstrate the refusal.

Free Gradio Spaces now run on ZeroGPU, which attaches a GPU only while a request is being processed. On this Space the ZeroGPU worker failed to initialise CUDA ("No CUDA GPUs are

available") with both PyTorch 2.13 and 2.9.1, before any of the app's code ran. I added an option, set through a Space variable, that runs the model on the Space's own CPU in fp32 instead, and with it the app answers quickly enough for interactive use. The code for the app is in the app folder of the repository.

9. Limitations and Future Work

9.1 Limitations

AfriMed-QA is English only, so the system cannot serve users who prefer other African languages. Half of the rows (7,577 of 15,275) come from Nigeria, which skews the training data toward Nigerian practice and phrasing. Only 7.6% of MCQs have explanations, so most MCQ training targets teach only a letter, and the consumer queries have no answers, which is why the model is weakest on the lay-language questions that real users would ask. My re-split also means the results cannot be compared directly with published AfriMed-QA scores.

A 1.5B model fine-tuned on 3,663 examples still answers about 45% of held-out MCQs wrongly, hallucinates in half of the reviewed answers and occasionally falls into repetition loops.

ROUGE-L and BERTScore measure similarity to one reference and reward matching length. The human ratings were done by one person with no medical training, so I could not compute inter-rater agreement and some judgements may be wrong. The challenge set has only 60 questions, I wrote them myself, and they informed the version 2 design. Each experiment ran once with one random seed, so small differences such as E2 against E3 should not be over-read.

The deployed system has no safety guarantees. The domain classifier lets through 43% of the tricky non-medical questions, its 0.5 threshold was not tuned on held-out data, and the generator gives no sources for its claims.

9.2 Future work

With more compute, the same recipe could be applied to a 7B model, since size was the factor that mattered most here. Training with retrieved passages in the prompt would give retrieval a fair test and could let the system show users where an answer comes from. Training for one or two epochs with early stopping would save time without losing accuracy. The evaluation would be stronger with clinician raters, a second rater and Cohen's kappa, and several training seeds. Answers for consumer queries written or checked by clinicians would address the model's weakest area. For the classifier, adding non-medical questions that use medical words, and choosing the threshold on a separate validation set, would target its main failure.

10. Conclusion

ClinicMate fine-tunes Qwen2.5-1.5B-Instruct with QLoRA on AfriMed-QA, puts a DistilBERT classifier in front of it to refuse non-medical questions, and runs as a free web app. Fine-tuning raised MCQ accuracy from 49.7% to 54.1% and made short answers match the style of the references, and the deployed merged model reached 54.9%. Model size mattered much more than LoRA rank, and retrieval added at inference time helped SAQ scores but hurt MCQ accuracy. Training on the crowdsourced official split barely transferred to expert questions.

The project taught me that automatic metrics can show large gains that hand review does not confirm: SAQ scores roughly doubled, yet half of the answers I reviewed contained a hallucinated claim and five were potentially harmful. It also showed that a domain classifier can look perfect on its own test

split while learning the wrong signal, and that a small hand-written challenge set is enough to expose this. As a research prototype, ClinicMate shows what a 1.5B model can learn from AfriMed-QA on free hardware, and its error rate means it should not be used for medical advice.

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